4 Moist thermodynamics: Phase diagrams, Clausius-Clapeyron relation, moist pseudo-adiabat

Reading for Day 4: Pierrehumbert Principles of Planetary Climate, Section 2.6, 2.7.

Most atmospheres in the Solar System (and presumably beyond) have species that are *condensible* – that is, they can condense (change phase from gaseous to liquid) or undergo deposition (change of phase from gaseous to solid). We will use the more general term "condense" often throughout this course to refer to both processes. Condensibles in the Solar System include water vapour (on Earth, Mars, and the gas and ice giants), carbon dioxide (Mars), methane (Titan and the ice giants), nitrogen (Triton, Pluto), ammonia (gas and ice giants) and hydrogen sulphide (gas and ice giants). Note that condensation is a local process, and typically occurs because air parcels undergo changes in their local temperature and pressure either due to radiative processes or atmospheric motion that lead them to reach a phase boundary where condensation occurs.

4.1 Phase diagrams

Phase diagrams are used to delineate where in temperature-pressure space a given substance is stable as a gas, liquid, or solid. Figure 4.1 shows a generalised phase diagram for an atmospheric species. The phase boundaries are clear between gas (stable at high temperature, low pressure), liquid (stable at high temperature, high pressure), and solid (stable at low temperature, high pressure). There are also two key locations on the phase diagram: 1) the triple point, which is where solid, liquid, and gas can co-exist[18], and 2) the critical point, which is where the distinction between liquid and gas disappears[19]. See Figure 2.4 for critical

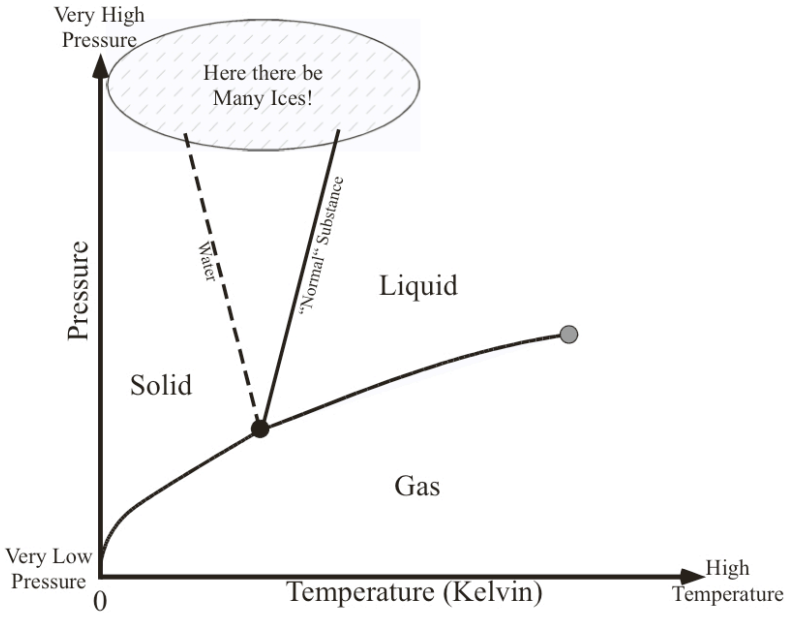


Figure 4.1: A general schematic form of a phase diagram (Figure 2.4 of Pierrehumbert, 2010). The triple point is a black circle, the critical point is a grey circle, the solid-liquid phase boundary for a normal substance (for which the solid phase is denser than its liquid phase) is shown as a solid curve, and the phase boundary for water (not a normal substance by this definition) is shown as a dashed curve.

[18]Notably, the temperature of Mars' atmosphere is close to the triple point of CO_2.

[19]Supercritical species beyond the critical point typically become miscible with other species, which is thought to prevent discrete layering of the interiors of gas and ice giant planets.

and triple points of various atmospheric species – for water, the critical point occurs at a temperature of 647.1 K and pressure of 221 bars, clearly far beyond the conditions seen on Earth.

4.1.1 Latent heat

Latent heat (typically denoted L) is the amount of energy released when a unit of mass of a substance changes phase while holding temperature constant. Thus,

$$Q = mL, \tag{4.1}$$

where Q is the amount of energy released and m is the mass of a substance. The *latent heat of vaporization* (also sometimes termed the latent heat of condensation when in the reverse direction) corresponds to the heat required to vaporise a substance from liquid to gas (or the heat output when a substance condenses from gas to liquid). The *latent heat of fusion* corresponds to the heat required to change the phase of a substance from solid to liquid (or the heat output when a substance changes phase from liquid to solid). The *latent heat of sublimation* corresponds to the heat required to change the phase of a substance from solid to gas (or the heat output when a substance changes phase from gas to solid). Note that the sign convention is dependent on the direction of heat transfer, with positive referring to energy released and negative referring to energy input.

The latent heat can often reasonably be assumed to be constant over a broad range of temperature, but they do decrease with increasing temperature and of course go toward zero at the critical point.

Question for the class: Why does latent heat decrease toward zero when approaching the critical point?

For example, the latent heat of vaporisation of water decreases from 2.493×10^6 J kg^{-1} at the triple point (273.15 K) to 2.255×10^6 J kg^{-1} at the critical point (647.1 K), a $\sim$ 9.5% change – not large, but also not negligible. We will assume that the latent heat is approximately constant in the derivations that follow, but this should be relaxed for practical applications.

4.1.2 Activity – application of latent heat and phase diagrams: two thought experiments

Please get in groups of two or three to work on the following activity considering two thought experiments – the first is a direct calculation, and the second more of an estimation that has been conducted more rigorously in recent work.

1. Eventually as the Sun brightens on the main-sequence, Earth will undergo a runaway greenhouse (similar to Venus) where its oceans will boil and vaporise. Earth's oceans are approximately 0.023% of the total mass of the Earth (which is $M_\oplus = 5.97 \times 10^{24}$ kg). How long would it take Earth's oceans to boil, once their temperature reaches the boiling point of water? Assume for simplicity in this estimate that the received Solar flux on the surface (after taking into account albedo) is $\approx$ 1360 W m^2 (of course it will be a bit higher due to the main-sequence brightening, so the timescale is an upper limit).

2. The sub-Neptune K2-18b has a mass of $7.2M_\oplus$ and a radius of $2.5R_\oplus$, implying a bulk density of 2.28 g cm^{-3}. It receives a similar instellation to Earth, with an equilibrium temperature of ≈ 270 K. Its atmosphere is expected to potentially have $\gtrsim 10\%$ H_2O by volume. Assuming for simplicity that the atmosphere of K2-18b is comprised only of water (i.e., a steam atmosphere), would you expect this steam atmosphere to sit atop 1) an ocean, with a discrete atmosphere/ocean interface, or would it 2) be a totally fluid planet like Uranus and Neptune?

4.2 Clausius-Clapeyron relation

Clouds can form in planetary atmospheres where species are thermodynamically favoured to undergo a phase transition from vapour to solid phases. Cloud condensation is thermodynamically allowed when the partial pressure of a species exceeds its saturation vapour pressure. Critically, the saturation vapour pressure is extremely sensitive to temperature, as we discuss below.

The saturation vapour pressure is set by the Clausius-Clapeyron relationship[20], which for an ideal gas can be expressed as

$$\frac{dp_{\rm sat}}{dT} = \frac{1}{T}\frac{L}{\rho_{\rm vap}^{-1} - \rho_{\rm cond}^{-1}}, \tag{4.2}$$

where $\rho_{\rm vap}$ is the density of the vapour phase, $\rho_{\rm cond}$ is the density of the condensed phase, and $p_{\rm sat}$ is the saturation vapour pressure. If $\rho_{\rm cond} \gg \rho_{\rm vap}$ (as is the case for most relevant atmospheric condensibles), then (using the ideal gas law to relate $\rho_{\rm vap}$ to p) Clausius-Clapeyron simplifies to

$$\frac{dp_{\rm sat}}{dT} \approx \frac{p_{\rm sat}L}{RT^2}, \tag{4.3}$$

where $p_{\rm sat}$ is the saturation vapour pressure and L is the latent heat of vaporisation/condensation, fusion, or sublimation of the species of interest. If we assume $L \approx$ constant, Equation (4.3) can be directly integrated to obtain the saturation vapour pressure curve

$$p_{\rm sat} = p_0 \exp\left[\frac{L}{R}\left(\frac{1}{T_0} - \frac{1}{T}\right)\right], \tag{4.4}$$

implying that the saturation vapour pressure scales exponentially with $-1/T$. As a result, the saturation vapour pressure of a given species decreases with decreasing temperature, as shown in Figure 4.2. Note that this decrease in saturation vapour pressure with decreasing temperature occurs for all species, but the exact slope depends on the condensible of interest. Figure 4.3 shows the saturation vapour pressure as a function of temperature for water vapour in particular. Given the steep decline of temperature with height for an adiabatic atmosphere (derived earlier in Section 3.1), this implies that condensible water vapour mixing ratios are orders of magnitude smaller at the tropopause than at the surface. It also more generally implies that condensible water vapour mixing ratios are smaller in colder regions of the

[20]For those of you who completed their first three years at Oxford, you may recall deriving this in A1.

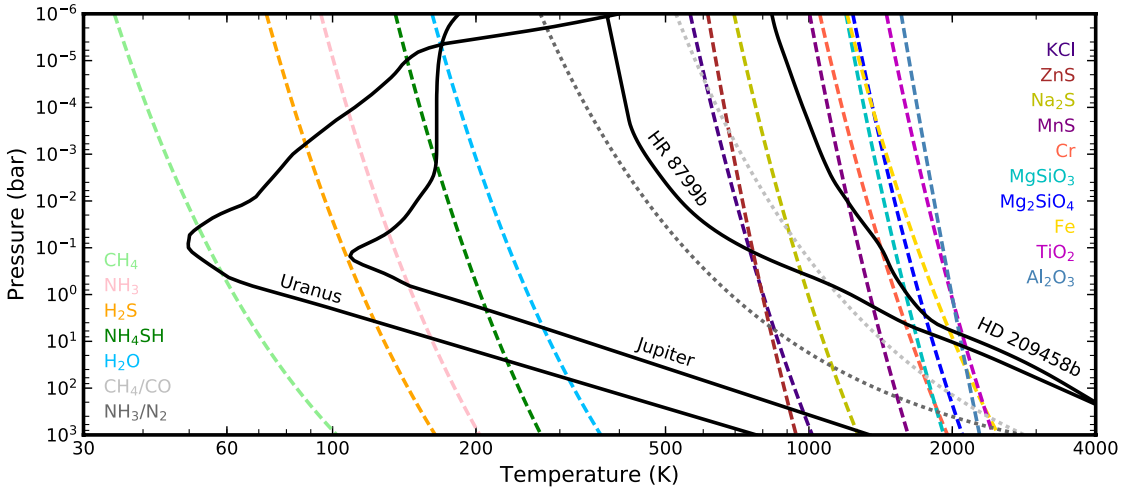


Figure 4.2: Temperature profiles of Solar System giant planets and exoplanets (black lines) compared with condensation curves of various species (dashed lines) and equilibrium chemistry equivalency curves (dotted lines). Figure adapted from Gao et al. (2021).

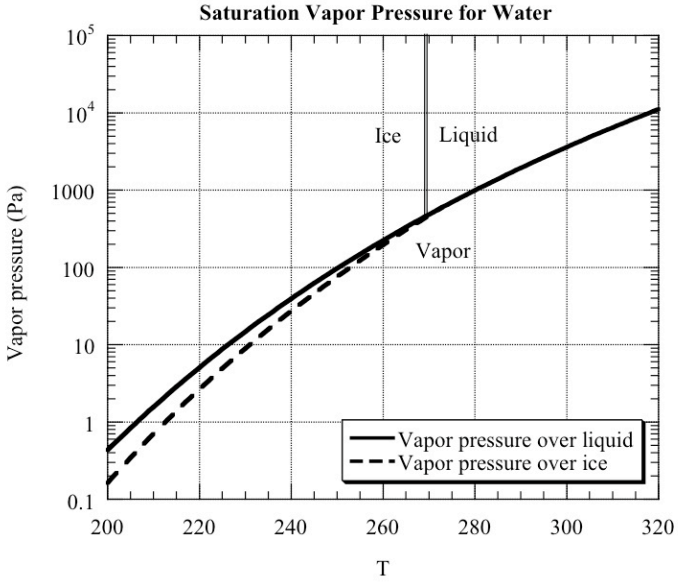


Figure 4.3: Saturation vapour pressure as a function of temperature for water vapour. Figure adapted from Pierrehumbert (2010)

planet, e.g., implying a decrease in atmospheric moisture from the hotter equator toward the colder pole[21].

Using condensation temperature instead of saturation vapour pressure, one can equivalently state that condensation from gas to cloud (solid or liquid) is thermodynamically allowed at a given pressure when $T < T_{\rm cond}$, which then occurs when temperature profiles cross the condensation curves in Figure 4.2[22]. This is why clouds generally form at high

[21]The low saturation vapour pressure for cold temperatures is why, for instance, one's skin can become relatively dry in the winter (this is even more of an issue for those in colder, drier places like Toronto, Boston, or Chicago than the relatively humid Oxford).

[22]As a planetary scientist, I personally prefer using condensation temperature over saturation vapour pressure as a rule-of-thumb because the variation in temperature between planets is typically much larger

altitudes in planetary atmospheres – because the temperature decreases with height in the troposphere, clouds form at altitudes where the temperature is cool enough to allow for condensation. The exception is when the local saturation vapour pressure is enhanced (e.g., due to mixing), increasing the partial pressure of a species to allow for condensation.

4.3 Single-component moist (pseudo)-adiabat

The simplest case of a condensible atmosphere is one where the entire atmosphere is made up of the condensible species. There are no perfect examples of this in the Solar System, but Mars (CO_2-dominated) and Triton (N_2-dominated) have atmospheres that can be well approximated as having a single condensible, as likely can the atmospheres of some exoplanets (e.g., super-Earth or sub-Neptune waterworlds with steam-rich atmospheres). In this case, the saturation vapour pressure is then the total atmospheric pressure when condensation occurs. This implies that the temperature gradient we previously derived in Equation (4.3) from Clausius-Clapeyron is then simply the moist-adiabatic lapse rate of the atmosphere. More generally, note that this is termed the moist *pseudo*-adiabat rather than the moist adiabat – I will call it moist adiabat as shorthand, but this is slightly incorrect. The reason that this is a pseudo-adiabat rather than an adiabat is that we assume the condensate is instantaneously removed from the parcel, causing the process of condensation to not be reversible since the condensate mass has left the parcel. The moist pseudo-adiabat is approximately the true moist adiabat if the amount of retained condensate mass is small (i.e., if "rainout," that is the removal of condensate from the vapour, is a reasonable approximation), which is generally true for Earth.

For an atmosphere composed of a single condensible[23] component, the Clausius-Clapeyron relationship can be expressed as a single-component moist adiabatic lapse rate

$$\frac{d\ln T}{d\ln p} = \frac{RT}{L}, \tag{4.5}$$

where p here is both the total pressure and the saturation vapour pressure (because the gas species is always at saturation if it is the only vapour component).

Given that the dry adiabatic gradient is $d\ln T/d\ln p = R/c_p$, the moist adiabatic gradient is smaller than the dry adiabatic gradient if $L > c_p T$ – which is almost always satisfied for most condensible species of interest, including water, carbon dioxide, methane, and nitrogen (which typically have $L/(c_p T) \approx 2 - 6$, see Figure 2.4).

Note also that in the single-component limit if there is no rainout of condensate (i.e., condensate remains in the air parcel being considered), then moist-adiabatic perturbations to temperature are reversible. This means that if condensation of vapour occurs as air rises, then vaporisation/sublimation of condensate occurs as it sinks, and the resulting air parcel follows the moist-adiabatic lapse rate for both ascent and descent. However, in the real case of Earth this is incorrect, both because it is not comprised of a single-component atmosphere and because descending air in the mid-latitudes is typically drier than ascending air in the hot, moist tropics due to rainout of condensed water from air in the moist regions (see also

than the variation in the atmospheric pressure ranges of interest.

[23] Any gaseous species is a condensible if one considers cold enough temperatures! Even H_2 (which has a condensation temperature of 20.28 K).

our caveat above r.e. the "pseudo"-adiabat). In the case where descent occurs after the air has dried out, then the descending air follows the steeper dry adiabat[24].

Finally, note that in the case where $L \approx$ constant, we can invert our previously derived expression for the saturation vapour pressure as a function of temperature (Equation 4.4) to derive the single-component moist adiabat:

$$T(p) = \frac{T_0}{1 - \frac{RT_0}{L} \ln\left(\frac{p}{p_0}\right)}. \tag{4.6}$$

Note that Equation (4.6) only governs the entire atmospheric temperature profile if the temperature at the surface is cold enough for condensation to occur all the way to the ground (an example of this is the pole of the winter hemisphere on Mars). In warmer conditions (e.g., the low latitudes of Mars), then the near-surface temperatures are too warm for condensation, and the near-surface atmosphere follows the dry adiabat. The atmospheric lapse rate then only follows the moist adiabatic lapse rate above the condensation level at which saturation finally occurs – see Figure 4.4 for an example applicable to the warmer regions of present-day Mars. In this case, the atmosphere follows a dry adiabat from the surface (at

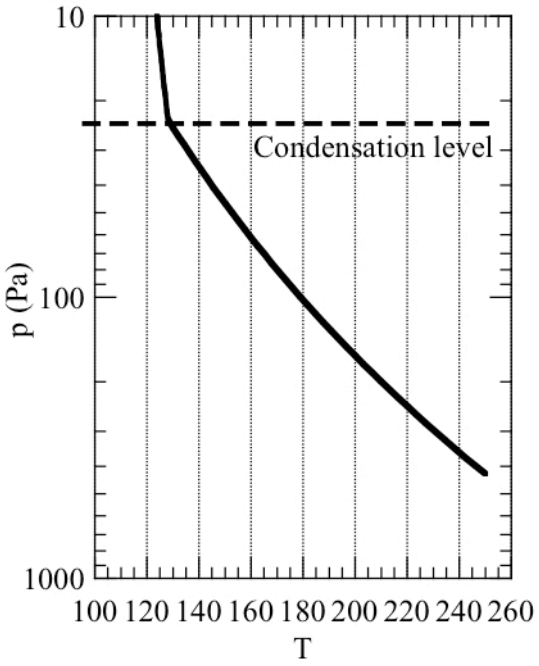


Figure 4.4: Adiabatic profile for a pure CO_2 atmosphere with a surface pressure and temperature similar to warm regions on present-day Mars. Figure adapted from Pierrehumbert (2010).

4.5 mbars) to the condensation level (0.25 mbars), above which it follows a moist adiabat. Note that as we expected, the moist adiabatic profile is shallower than the dry adiabat due to the latent heat of CO_2 exceeding the enthalpy of the dry CO_2 vapour.

[24]This is why Earth's sub-tropics near the descending branch of the Hadley cell are hot arid regions such as the Saharan and Sonoran deserts.